

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0601	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	V. Prabhavathi	Reg. No.:	192472067
Section / Batch:	CSE-AI	Date:	20/07/26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:			100 Marks

SECTION A – APPLICATION BASED QUESTIONS

48
50 ✓

Code of Conduct

I, G. Prabhavathi (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Prabhavathi

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)	20	28	20	20	10	48
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: <u>[Signature]</u>	Signature: _____	Signature: _____
Date: <u>20/7/24</u>	Date: _____	Date: _____

Col - Assessment Tool - 1
Application - Based

U. Prabhavathi
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CSA0601

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Q16 Application: Autonomous vehicle sensor system
A vehicle system processes sensor data using the recurrence $T(n) = 5T(n/2) + n^2$. Apply the Master's Theorem to determine the time complexity. Identify Parameters and determine the space complexity.

Autonomous Vehicle Sensor system

An autonomous vehicle sensor system is one of the most important components of a self-driving car. It uses sensors such as LIDAR, Radar, GPS, ultrasonic sensors and cameras to collect information.

Given Recurrence Relation

$$T(n) = 5T(n/2) + n^2$$

$T(n) \rightarrow$ Time required to process

$5T(n/2) \rightarrow$ The problem is divided

$n^2 \rightarrow$ Time required to combine

Master Theorem Parameters

$$T(n) = aT(n/b) + f(n) \Rightarrow a=5, b=2, f(n)=n^2$$

Calculate $n \log_b a$

$$n \log_2 5 \text{ Since } \log_2 5 \approx 2.32$$

$$\Rightarrow n \log_2 5 = n^{2.32}$$

Time complexity

$$T(n) = O(n \log_2 5)$$

$\boxed{O(n^{2.32})} \Rightarrow$ Big O notation:

$$\boxed{O(n^{2.32})}$$

Order of Growth

The algorithm grows approximately as $O(n^{2.32})$
→ It is slower than $O(n^2)$ but faster than $O(n^3)$

Efficiency Analysis

The processes massive sensor data continuously. The divide-and-conquer strategy allows multiple sensor data blocks to be processed simultaneously.

Space complexity

$$O(\log n)$$

For Master Theorem recurrence relations, cases are not applicable for this.

Q97 Applications: Distributed Database Merge system
A database system processes records using the recurrence $T(n) = 2T(n/2) + \log n$. Apply the Master's Theorem.

A distributed Database Merge system is used to combine data stored across multiple database servers into a single, consistent database.

Recurrence Relation

$$T(n) = 2T(n/2) + \log n$$

Master's Theorem:-

$$T(n) = aT(n/b) + f(n), \quad a=2, b=2, f(n) = \log n$$

Calculate:- $n \log_b^a$

$$n \log_2^2 = n, \quad \text{so } n \log_b^a = n$$

compare $f(n)$ and $n \log_b^a$

$$f(n) = \log n$$

$$\log n = O(n^{1-\epsilon})$$

for some $\epsilon > 0$

the function $f(n)$ is polynomially smaller than n .

Time Complexity

$$T(n) = O(n)$$

Time complexity (order of growth) :- $O(n)$

Efficiency Analysis

The Distributed Database Merge system is highly efficient because it divides the database into two equal parts and processes them simultaneously.

Space Complexity

The recursion depth is $\log_2 n$

Hence, space complexity

$$O(\log n)$$

⑤ This means the memory required grows only slowly with the input size.
order of growth: $O(n)$

898 Application: Video Streaming Quality Adjustment System

A streaming system processes frames using the recurrence $T(n) = 2T(n/2) + 1$. Apply Master's theorem.

A Video Streaming Quality Adjustment System is used by streaming platforms such as YouTube, Netflix, and Amazon Prime Video to provide smooth video playback.

Recurrence Relation:-

$$T(n) = 2T(n/2) + 1$$

Master's Theorem:-

$$T(n) = aT(n/b) + f(n), \quad a=2, b=2, f(n)=1$$

Calculate $n \log_b a$

$$n \log_2 2 = n, \text{ so } n \log_b a = n$$

Compare $f(n)$ and $n \log_b a$

$$f(n) = 1$$

$$1 = O(n^{-\epsilon})$$

for some $\epsilon > 0$

the function $f(n)$ is Polynomically smaller than n

Time complexity :-

$$T(n) = O(n)$$

$$\Rightarrow O(n)$$

Space complexity

$$\log_2 n$$

$$\Rightarrow O(\log n)$$

order of growth :-

$$O(n)$$

Q99

Application :- E-commerce order Matching System
An order matching system processes data using the recurrence $T(n) = 3T(n/2) + n \log n$ Apply

Masters Theorem:

An E-commerce order Matching system is used by online shopping platforms as Amazon, Flipkart, and eBay to match customer orders with available products in warehouses.

Recurrence Relation

$$T(n) = 3T(n/2) + n \log n$$

Masters Theorem

$$T(n) = aT(n/b) + f(n), a=3, b=2, f(n)=n \log n$$

calculate $n \log_b a$

$$n \log_2 3$$

$$\log_2 3 \approx 1.585$$

$$n \log_2 3 = n \cdot 1.585$$

compare $f(n)$ and $n \log b^a$

$$f(n) = n \log n$$

$$n \log n = O(n^{1.585 - \epsilon})$$

for some $\epsilon > 0$
the function $f(n)$ is Polynomially smaller than $n^{1.585}$

Time complexity

$$T(n) = O(n \log_2 3)$$

$$O(n^{1.585}) \Rightarrow O(n^{1.585}) \text{ Big-O notation.}$$

Order of growth

$$O(n^{1.585})$$

faster than $O(n^2)$, slower than $O(n \log n)$

Efficiency Analysis

The E-commerce order matching system efficiently processes customers orders using a divide-and-conquer strategy.

Space complexity

The recursion depth is $\log_2 n$

$$O(\log n)$$

Q100 Application: cybersecurity Intrusion Detection system

A cybersecurity system processes logs using the recurrence $T(n) = T(n/2) + n \log n$. Solve the recurrence using an appropriate method to determine time complexity.

A cybersecurity Intrusion Detection system is a security application used to monitor computer networks and detect unauthorized access, malware, viruses, ransomware, and other cyberattacks.

Recurrence Relation

$$T(n) = T(n/2) + n \log n$$

Identify the Parameters

$$T(n) = aT(n/b) + f(n) \quad a=1, b=2, f(n)=n \log n$$

calculate $n \log_b a$

$$n \log_2 1 = n^0 = 1$$

Compare $f(n)$ and $n \log_b a$

$$f(n) = n \log n$$

$$n \log n = \Omega(n^{\epsilon})$$

for some $\epsilon > 0$ and the regularity condition is satisfied.

Time Complexity

$$T(n) = \Theta(f(n))$$

$$T(n) = \Theta(n \log n)$$

$$\boxed{\Theta(n \log n)} \text{ Big } \Theta \text{ notation}$$

order of growth

$$\boxed{\Theta(n \log n)}$$

Efficiency Analysis

The cybersecurity Intrusion Detection System efficiently analyzes network logs using a divide and conquer strategy.

Space Complexity

$$\boxed{\Theta(\log n)}$$

Best case: $\Theta(n \log n)$

Average case: $\Theta(n \log n)$

Worst case: $\Theta(n \log n)$